Silent Evidence Defects in the Wild: A Registered Replay of a Real Purchase-to-Pay Process

Gaston Besanson*

Abstract

Of the 155,534 analysed invoice clearings of a real purchase-to-pay event log (Section 4 gives the registered exclusions), 17,748 (11.41%, clustered bootstrap 95% interval [10.94%, 11.92%]) were made on evidence that failed at least one of seven registered predicates, carrying 19.41% of the cleared value ([16.04%, 23.09%]). We replay every analysed clearing of the BPI Challenge 2019 log through a deterministic min() gate declared before the data were read, over the evidence reconstructed from the log at clearing time. Clearings made on evidence that failed on the goods receipt (absent or late, D1/D2) were followed by a corrective event within 60 days at a rate of 5.64%, against 0.93% for three-way clearings with every applicable predicate passing (difference 0.0470, Holm-adjusted $p = 4.53 \times 10^{-4}$); clearings made after a change of price or quantity since approval (D4 fail), at 1.91% against 0.96% for D4-pass clearings (difference 0.0094, Holm-adjusted $p = 3.48 \times 10^{-7}$). On a registered sample of lines with a failed predicate and of clean lines, two model families shown the ERP view held 100.0% and 99.6% of the first and 53.0% and 57.3% of the second; on the sample lines the log lets us classify, people held 33.0% and 34.0%, where a hold is a recorded payment block or, in nearly every case, an inferred non-release within 60 days, not a recorded decision; the gate's verdict holds the first and allows the second by construction. A line's status at the decision point follows a reading of D5 (decision D-C-2 B) that the author fixed after E3 and before any model call (Section 4). Paper 5's compiler, given the three declared losses of the registration, does not reproduce the registered contract in any item category. The coverage matrix states which defect classes this log cannot expose; the authority defect class (D5) has zero incidence here. No causal claim is made.

1 Introduction

Earlier papers of this programme showed the mechanism of a silent evidence defect, a decision taken on evidence that fails a metadata-borne predicate, on constructed domains, and made no prevalence claim. The six papers are on arXiv: paper 1, SARC, on whether the agent may act (Besanson 2026a); paper 2, Green SARC, on its budget (Besanson 2026b); paper 3, SARC-DQ, on the evidence it admits (Besanson 2026c); paper 4 on composing several gates on one action (Besanson 2026d); paper 5 on deriving the contract, the compiler (Besanson 2026e); and paper 6 on facts read from documents (Besanson 2026f). This paper is the number, on one real process.

The unit is an invoice clearing in the BPI Challenge 2019 purchase-to-pay log. For each of the 155,534 analysed clearings (the exclusion table of Section 4) we reconstruct the evidence recorded on the purchase-order line at clearing time, evaluate the registered predicates D1 to D7, record what the gate would have held, and read what followed in the log. We report exposure in count and in anonymised EUR, the corrective events that follow a failed predicate, the reach of the gate, and, on a registered sample, what two model families decide on the same ERP view beside the people in the log and the gate. Everything was registered before the log was parsed beyond its published counts (tag `prereg-p7-v1`) and amended once after the pilot (v1.1, tag `prereg-p7-v1.1`), with five deviations under v1.1: two decided after E2 and E3 and before E4, and three found in review (corrected descriptions in the coverage matrix, the E7 bootstrap replicates with an empty group discarded, and the request format of the Jev family in E4); all are listed in Section 4.

*Universidad Torcuato Di Tella

What this paper claims is in the claim table of Appendix B. What it does not claim is stated in the repository's claims file and repeated here verbatim:

- No causal claim. Correctives are a proxy, observational, confounded by vendor, category and practice.
- No loss figure. A corrected payment is not money gone; recovery is not in the log.
- No claim that the gate stops anything. Held lines go somewhere the log does not show.
- No claim about the company or any vendor. No company, subsidiary or vendor is named in any finding.
- No agent acting on the live system and no agent with tools.
- No readiness threshold.

"No claim about the company" means that no company, subsidiary or vendor is named or characterised; the measured process is the subject of every finding.

2 Related work

This paper does not introduce conformance checking, the three-way match or rework analysis. It replays each clearing as a gate decision over the evidence reconstructed from the event log at clearing time, measures how often that evidence was defective and what followed, and puts people, models and the gate on the same lines. The defect classes are the predicates of Section 3.

Conformance and compliance checking. Checking recorded behaviour against a normative model or rule set is established (van der Aalst 2016; Ly et al. 2015), and declarative constraints in the Declare family take a process model as the conjunction of its constraints (Pesic et al. 2007). The predicates D1 to D7 are compliance rules in this sense, and taking the decision as their conjunction is Declare's. Synthesising a sufficient contract from declared losses is paper 5 of this programme (Besanson 2026e); the compiler used in Section 5.5 is paper 5's. Detecting that one resource performed two activities that must be separated is a standard compliance monitoring functionality (Ly et al. 2015); predicate D5 is that check.

The BPI Challenge 2019 reports. The challenge posed the process owner's compliance questions on this very log, including clearing before goods receipt and the three-way match per item category (ICPM 2019). The submitted reports measured basic violation rates and are prior art for every basic rate reported here; among them Diba et al. 2019 quantify receipt-before-clearing violations with their value, and Porter et al. 2019 report compliance per item category both by number and by value. Measuring compliance by count and by value is therefore not claimed; what is claimed is the tabulation per clearing and per defect class, with clustered intervals. The three-way match itself is a textbook application control (PCAOB 2004); Section 5.5 asks whether paper 5's compiler re-derives it from declared losses.

Deviation and outcome. Relating a deviation to a later outcome in the same case is established: the extension of process mining to outcomes associated with deviations (Peleg et al. 2008), outcome-oriented predictive process monitoring (Teinemaa et al. 2019) and deviance mining (Nguyen et al. 2016). On this log, Porter et al. 2019 observe cancellations of invoice receipts long after clearing as a case-study observation. Corrective events as rework signals are established on this log too: Meyer zu Wickern et al. 2019 count change, cancel and delete activities as rework, and Badakhshan et al. 2019 list vendor debit memos among rework activities. The outcome proxy of Section 3 is a rework signal of this kind; H1 to H3 are registered conditional-consequence studies of the deviation-to-outcome shape.

Detectability. That some deviations leave no trace in an event log is established (Outmazgin and Soffer 2013). The registered per-class coverage matrix of Section 6 is claimed; the principle is not.

Deciders on the same cases. Putting different deciders on the same cases is established: a human reviewer, a model agent and learned rules on the same accounts (Palma et al. 2026), language models against audit baselines on the same ledgers (Kadir et al. 2025), and algorithms against bank managers (Harding and Vasconcelos 2022). None of these uses a deterministic contract declared before the data were read, or a human arm read from a process event log; Section 5.4 does both, with one qualification: the status of a line at the decision point follows a reading of D5 (decision D-C-2 B) that the author fixed after E3 and before any model call (Section 4).

The BPI Challenge 2020 reports. The 2020 challenge posed questions on the approval chain of travel

declarations (ICPM 2020b); two reports received awards (ICPM 2020a; Pufahl et al. 2020; Kas et al. 2020). They are prior art for the basic rates of the replication in Section 5.7.

3 Formal core

Lines, clearings and evidence. A line is one case of the log: one purchase document and item. A clearing is one `Clear Invoice` event on a line at time t_c. The evidence at t_c is every event on the line with timestamp at most t_c, excluding the clearing itself, as the registration defines "before t_c" (section 2 of `prereg-p7-v1`, unchanged in v1.1); events sharing a timestamp count as available, and counts rather than event order decide receipts, blocks and deletions, so the verdict does not depend on the order of events with equal timestamps (a property test of the replay engine).

Category and contract. The case flags select the contract, as the data record states the rules:

category	GR-based invoice verification	goods receipt	contract
three-way, invoice after receipt (`3-way`)	true	true	D1 to D7
three-way, invoice before receipt (`3-way`)	false	true	D1 to D7
two-way (`2-way`)	false	false	D3 to D7
consignment	any	any	excluded

Predicates. Each predicate takes the value pass, fail or unknown; unknown means a required input is missing. Each predicate checks one SARC-DQ dimension (the table). The contract as a whole is checked with paper 5's own sufficiency function over this log's observed states (E5, `results/e5/sufficiency.json`, Section 5.5): it is sufficient for the three declared losses in the two-way category, and in the two three-way categories for L1 and L3 but not for L2; it is minimal in no category.

id	name	dimension	fails when, on the line before t_c
D1	receipt present	presence	no uncancelled goods receipt (receipts minus receipt cancellations at most zero)
D2	receipt before clearing	freshness, order	no uncancelled receipt precedes t_c
D3	invoice present	presence	no invoice receipt
D4	unchanged since approval	validity	a price or quantity change after the last approval-bearing event
D5	distinct actors	authority	the actor of the clearing equals an actor of the earliest creation (unknown if either is missing)
D6	block respected	scope	payment blocks set minus blocks removed greater than zero
D7	not deleted	validity	deletions plus blocks minus reactivations of the line greater than zero

At t_c, D1 and D2 have the same truth value; they are kept apart because they carry two dimensions. For reporting only, a D1/D2 failure is assigned to a defect class by what the log records later: the stale receipt (D2 class) if an uncancelled receipt is recorded after t_c and by the observation end, the absent receipt (D1 class) otherwise.

Verdict. The verdict is min() over the category's contract with pass as one and fail or unknown as zero: the gate allows a clearing if and only if every applicable predicate passes, and otherwise would have held it. A clearing is defective if at least one applicable predicate fails; one whose only non-pass value is unknown is held but not defective, and is counted separately. "Defect" is used only for a failed predicate.

Human hold. A line is held by people if a payment block is recorded before its first clearing, or if no clearing is recorded within W days of its first invoice receipt; a clearing before the first invoice receipt counts as a release.

Outcome proxy and lag bands. O = 1 for a clearing if a registered corrective activity (an invoice receipt or goods receipt cancelled, a price or quantity changed, the line deleted, a vendor debit memo, a subsequent invoice) is recorded on the line within W after t_c; W = 60 days, with W = 30 days as a descriptive sensitivity. For a D1/D2 failure, the lag from t_c to the first uncancelled receipt falls in one of the registered bands [0, 1], (1, 7], (7, 30] days, over 30 days, or never.

Evidence set. One record per analysed clearing carries the line id (hashed in every export), category, t_c, every predicate input and value, the verdict, the defect classes, O at both windows, the cleared and line values, and the SHA-256 of the canonical serialisation of the line's events up to t_c.

4 Data and registration

Data. The primary log is the BPI Challenge 2019 record; the replication uses the two declaration logs of the BPI Challenge 2020 collection. All three are by Boudewijn van Dongen on 4TU.ResearchData; each file was checked against the hash the registration records.

record	DOI	licence	file	cases	events	SHA-256
BPI Challenge 2019	doi:10.4121/uuid:d06aff4b-79f0-45e6-8ec8-e19730c248f1	CC BY 4.0	`BPI_Challenge_2019.xes`	251,734	1,595,923	`af63bc687fc4152f2123b05c3af7772b37ef3fce2d3f67f812666c9e356baae7`
BPI Challenge 2020: International Declarations	doi:10.4121/uuid:2bbf8f6a-fc50-48eb-aa9e-c4ea5ef7e8c5	CC BY-NC 4.0	`International Declarations.xes.gz`	6,449	72,151	`b527c8d06255f4312c6ce8f36f7f4110243d84fd6ca48583cd9b80896c601494`
BPI Challenge 2020: Domestic Declarations	doi:10.4121/uuid:3f422315-ed9d-4882-891f-e180b5b4feb5	CC BY-NC 4.0	`DomesticDeclarations.xes.gz`	10,500	56,437	`0bc3312acf0b9ee5ae6f4ebe9df204005ef11665d62f2d21dd04995161d7532b`

Values in the 2019 log are anonymised by a linear translation that respects zero; we report them as anonymised EUR.

Window, observation end and frame. A line is in the window if its earliest `Create Purchase Order Item` event lies in calendar 2018. The observation end T_end is the latest timestamp in the last calendar month that carries at least 1,000 events; here T_end is 2019-01-18 and 9 later events are excluded. A line is framed if it survives the exclusions below and its first clearing, or its creation if it has none, lies at least W before T_end; the analysed clearings are every clearing on a framed line at least W before T_end, so every outcome window is complete. The lines drawn for the pilot (E0) are removed before any estimate. As registered, the `Item Category` string disagrees with the flag-derived category on 0 of 251,734 lines (section 3.1; `results/e1/category-crosscheck.json`); every analysed clearing took its value from the clearing event itself (155,534 clearings; 0 fell back to the invoice receipt and 0 to the line value, section 3.4; `results/e1/value-fallback.json`); and the seeded 1,000-record sample of the evidence set, identifiers hashed, is exported as `results/e1/evidence-sample.jsonl` (section 3.9).

step	lines
lines in the log	251,734
2.1 (1) no Create Purchase Order Item	0
2.1 (2) consignment	14,498
3.1 category undefined by the flags	0
2.1 (3) zero or missing line value	1,887

step	lines
2.1 (4) more than K = 12 clearings	78
2.1 (5) outside the window	376
2.1 (5) failing the frame condition	53,731
framed lines	181,164
E0 pilot lines removed	500
framed lines analysed	180,664
analysed clearings	155,534
events after T_end excluded	9

Registration. The registration was tagged `prereg-p7-v1` before any parse of the log beyond the published counts. The pilot fired registered stop conditions (the latest timestamp lay outside the registered range, and a mapped attribute is named differently in the log than on the record page); the registration was amended and tagged `prereg-p7-v1.1`, and the pilot was drawn again, before any estimate. Every deviation, with the decision that records its reason:

- p7-v1 to p7-v1.1: T_end rule.
- p7-v1 to p7-v1.1: Mapping attribute name.
- p7-v1 to p7-v1.1: Section 3.6, explicit sentence.
- Under p7-v1.1: H3 convention.
- Under p7-v1.1: D5 applicability at a non-clearing decision point.
- Under p7-v1.1: Description corrections in COVERAGE.md, no predicate changes.
- Under p7-v1.1: E7 bootstrap replicates with an empty group discarded.
- Under p7-v1.1: E4 Jev request format.

The H3 convention and the D5 reading were the author's decisions after E2 and E3 and before any model call (decision D-C-2 in the decision log). The response logs date the calls: D-C-2 was committed at 2026-10-25 18:21:56 UTC (`7aea13f`) and the E4 sample drawn at 2026-10-25 18:39:34 UTC (`5ffd887`); the first request was sent at 2026-10-25 18:42:18 UTC for Jev and at 2026-10-25 18:46:32 UTC for Haiku (`results/e4/timeline.json`; request times as the client wrote them before sending, commit times as git records them). The prior-art fence was reviewed externally after the tag `prereg-p7-v1` and before `prereg-p7-v1.1`, after two machine sweeps whose output is committed unedited.

5 Results

5.1 Exposure (E1)

defect class	clearings	share	Wilson 95%	bootstrap 95%	share of value	bootstrap 95%
D1 class (absent receipt)	610	0.39%	[0.36%, 0.42%]	[0.30%, 0.50%]	0.12%	[0.08%, 0.17%]
D2 class (stale receipt)	82	0.05%	[0.04%, 0.07%]	[0.02%, 0.10%]	0.12%	[0.03%, 0.27%]
D3	362	0.23%	[0.21%, 0.26%]	[0.15%, 0.33%]	0.03%	[0.02%, 0.05%]
D4	17,157	11.03%	[10.88%, 11.19%]	[10.55%, 11.53%]	19.24%	[15.88%, 22.89%]
D5	0	0.00%	[0.00%, 0.00%]	[0.00%, 0.00%]	0.00%	[0.00%, 0.00%]
D6	4	0.00%	[0.00%, 0.01%]	[0.00%, 0.01%]	0.03%	[0.00%, 0.10%]
D7	19	0.01%	[0.01%, 0.02%]	[0.01%, 0.02%]	0.01%	[0.00%, 0.02%]
at least one defect	17,748	11.41%	[11.25%, 11.57%]	[10.94%, 11.92%]	19.41%	[16.04%, 23.09%]
held on an unknown only (not a defect)	0	0.00%	[0.00%, 0.00%]	[0.00%, 0.00%]	0.00%	[0.00%, 0.00%]

Figure 1 shows the shares with their intervals. Shares of clearings carry a Wilson and a purchase-document-clustered bootstrap 95% interval (B = 10,000, seed 20261201); shares of value carry the clustered interval. The denominator is every analysed clearing, and a share counts the clearings made on evidence that failed the

predicate; D1 and D2 classes apply to 155,346 clearings in the three-way categories. A change after approval (D4) is the largest defect class, 17,157 clearings and 19.24% of cleared value; no clearing fails D5. Appendix D gives the same shares per defect class, with the clustered intervals, per item category, spend area, month, line value quartile and vendor activity quartile, reported beside the estimates and never modelled in. Section 4 reports the registered category cross-check, the cleared-value sources and the exported sample.

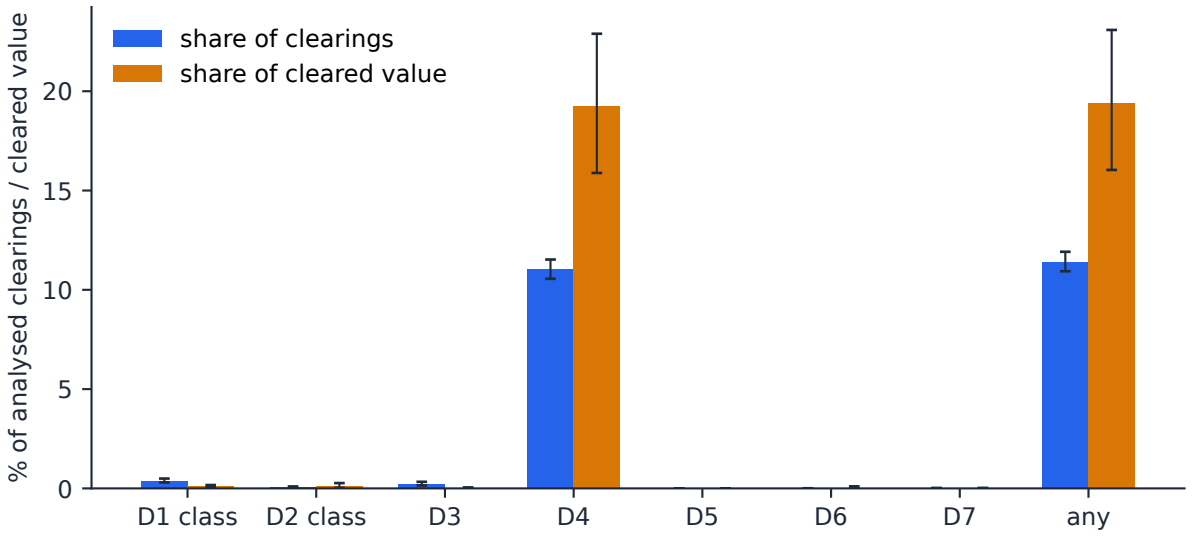


Figure 1: Figure 1: share of analysed clearings and of cleared value per defect class, with clustered bootstrap 95% intervals.

5.2 Consequence (E2)

test	defect group (failed predicate): O rate	comparison: O rate	difference	bootstrap SE	z	p (raw, one-sided)	p (Holm)
H1	39 of 692 (5.64%)	1,286 of 137,602 (0.93%)	0.0470	0.0127	3.688	1.13×10^{-4}	4.53×10^{-4}
H2	327 of 17,157 (1.91%)	1,332 of 138,377 (0.96%)	0.0094	0.0018	5.266	6.96×10^{-8}	3.48×10^{-7}
H3	0 of 0 (n/a)	1,659 of 155,534 (1.07%)	n/a	n/a	n/a	not computable	1

O is the registered corrective within W = 60 days. The H1 comparison group is three-way clearings with every applicable predicate passing; against clearings of every category with every applicable predicate passing the difference is 0.0470, descriptively. H2 compares D4 fail with D4 pass in every category. Standard errors come from the purchase-document-clustered bootstrap. Holm's adjustment runs over the registered family of five tests, H1, H2, H3 and H4 for each model family, at alpha = 0.05:

test	p (raw)	rank	multiplier	p (Holm)	below alpha
H2	6.96×10^{-8}	1	5	3.48×10^{-7}	yes
H1	1.13×10^{-4}	2	4	4.53×10^{-4}	yes
H3	1	3	3	1	no

test	p (raw)	rank	multiplier	p (Holm)	below alpha
H4-Haiku	1	4	2	1	no
H4-Jev	1	5	1	1	no

H3, distinct actors, is not computable: defect group has no analysed clearings. No analysed clearing was made by an actor of its line's earliest creation event, so the defect class has zero incidence; across the log the actors who record a `Clear Invoice` and those who record the earliest creation of a line are separate populations (23 and 185 actors, 0 in common; Section 6). No analysed clearing has D5 unknown (0): the 7,941 `Clear Invoice` events with no actor all lie outside the analysed set: of the 94 on framed lines, 94 fall after the horizon T_end - W (`results/e1/analysed-clearing-actors.json`). H3 enters the family with p = 1, the convention the author fixed in decision D-C-2, committed at 2026-10-25 18:21:56 UTC, before the first model request at 2026-10-25 18:42:18 UTC (`results/e4/timeline.json`). At W = 30 days the differences are 0.0354 for H1 and 0.0055 for H2, descriptively. Appendix E gives H1 and H2 per item category, spend area, value quartile, vendor activity quartile and month, descriptively and never modelled in.

5.3 Gate effect and hold cost (E3)

The gate would have held 17,748 of 155,534 analysed clearings (11.41%, 19.41% of cleared value); 0 of them were held on an unknown predicate value alone. Against the corrective at W = 60 days, in clearings and anonymised EUR (Figure 4):

gate verdict	corrective within W	no corrective within W
would have held	368 (21,418,557)	17,380 (145,068,897)
would have allowed	1,291 (106,294,821)	136,495 (584,977,346)

Appendix E gives the same table per stratum.

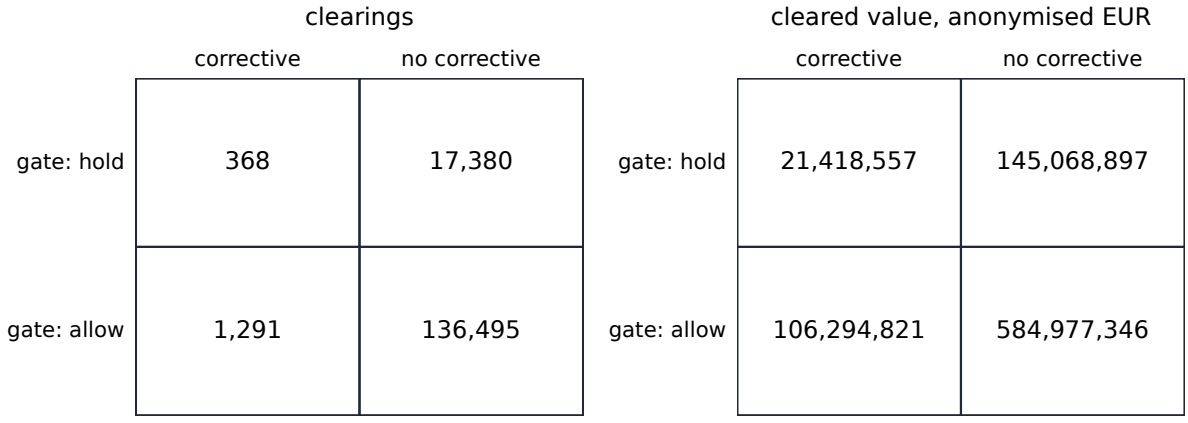


Figure 2: Figure 4: gate verdict against the corrective within W = 60 days, in clearings and in cleared value.

For D1/D2 failures, the time from clearing to the first uncancelled receipt (Figure 3):

lag band	clearings	cleared value
[0, 1] days	11	109,886
(1, 7] days	21	113,119
(7, 30] days	11	803,512
over 30 days	39	44,240
never (by T_end)	610	1,015,565

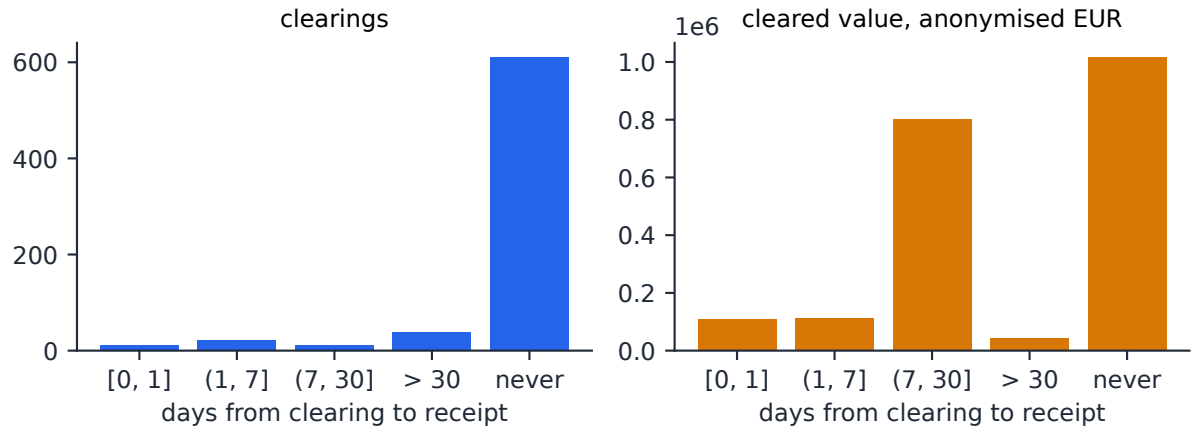


Figure 3: Figure 3: D1/D2 failures by lag band, in clearings and cleared value.

Human hold. Of 180,664 framed lines, people held 53,262 and released 101,777; 25,625 had no invoice receipt by T_end - W and are not classified. The status of a line is the gate verdict at the decision point t* of the agent replay: the first clearing if it came within W of the first invoice receipt, otherwise that receipt plus W. The registered text left D5 at a t* that is not a clearing undefined; read literally it is unknown, and then a line with no clearing within W can never be clean under the literal reading. The author amended the reading before any model call (at a t* that is not a clearing, D5 is not applicable: the decider there is the gate or a model; decision D-C-2 B); both readings are reported:

reading	predicate status at t*	classified lines	held by people	human hold rate
literal (registered text)	defective	16,264	5,664	34.83%
literal (registered text)	clean	91,178	1	0.00%
literal (registered text)	held unknown	47,597	47,597	100.00%
amended (D-C-2 B)	defective	16,264	5,664	34.83%
amended (D-C-2 B)	clean	138,775	47,598	34.30%
amended (D-C-2 B)	held unknown	0	0	n/a

Under the amended reading, people held 34.83% of lines with a failed predicate at t* and 34.30% of clean lines.

5.4 People, models and the gate on the same lines (E4)

Sample and arms. From the framed lines with an invoice receipt, 18,006 had at least one failed predicate at t* and 147,240 were clean, under D-C-2 B (`results/e4/sample-frame.json`). These differ from the 16,264 and 138,775 lines of Section 5.3 (`results/e3/human-hold.json`), which are the lines the people arm can classify: a line whose first invoice receipt comes later than T_end - W is eligible for the sample but not classified (section 3.6). The registered sample drew 1,000 defective and 1,000 clean lines, at most one per purchase document, stratified by category and value quartile, with seed 20261201; the draw was committed before any call. Each line was shown to two model families as the registered ERP view at t*, a short record with no event timeline and no company, vendor, actor or line id: TypeSafe System One (Jev, returned model `jev-1.14.0`) through `typesafe-sdk==0.7.2`, asked one choice question with labels clear and hold, and the Anthropic Messages API (`claude-haiku-4-5-20251001`, temperature zero, assistant prefill), asked for a JSON decision and hold probability. For Jev the template's closing reply-format sentence was replaced by

that typed choice, a deviation (Section 4); Haiku received the template as registered. Invalid replies, after one re-request: 0 for Jev and 0 for Haiku. The people arm is read from the log (Section 3); 152 sample lines have no invoice receipt by T_end - W and are not classified (section 3.6), so its denominators are smaller. The gate's verdict holds every line with a failed predicate and allows every clean line by construction.

arm	held, lines with a failed predicate	held, clean lines	kappa with gate	value released on lines with a failed predicate	expected correctives
people (from the log)	301 of 912 (33.0%)	318 of 936 (34.0%)	n/a	4,759,520	12.260
Jev	1,000 of 1,000 (100.0%)	530 of 1,000 (53.0%)	0.470	0	0.000
Haiku	996 of 1,000 (99.6%)	573 of 1,000 (57.3%)	0.423	7,307	0.076
gate	1,000 of 1,000 (100.0%)	0 of 1,000 (0.0%)	n/a	0	0.000

family	d (failed predicate minus clean)	SE	p, lower	p, upper	p_TOST
H4-Jev	0.470	0.0158	2.33×10^{-238}	1	1
H4-Haiku	0.423	0.0158	5.54×10^{-198}	1	1

Source: `results/e4/arms.json`, `results/e4/h4.json`.

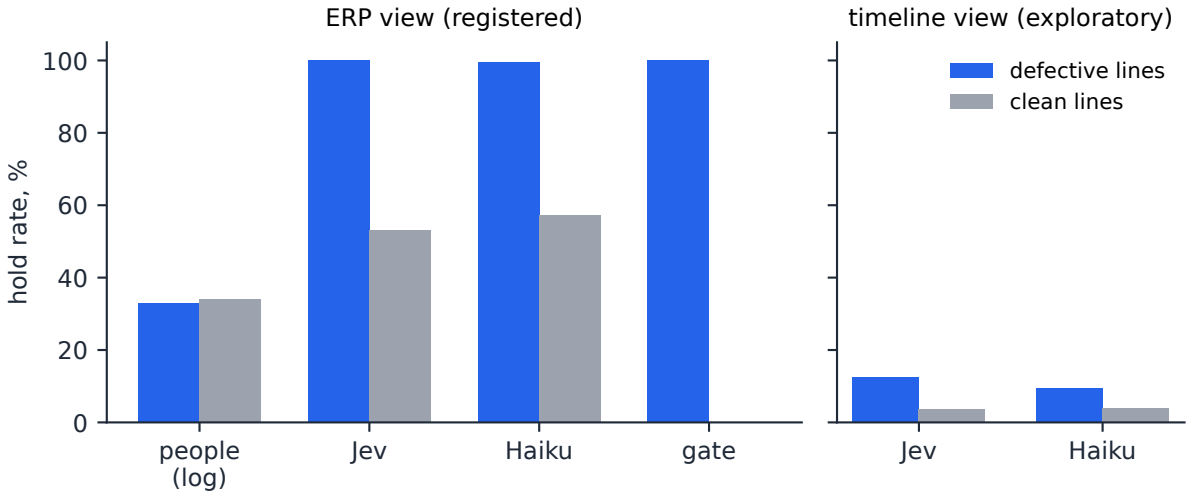


Figure 4: Figure 2: hold rate on lines with a failed predicate and on clean lines, per arm; the exploratory timeline arm in its own panel.

H4 asks whether each family's hold rate on defective lines (a failed predicate) is within m = 0.05 of its rate on clean lines. Neither is: Jev held 100.0% and 53.0%, Haiku held 99.6% and 57.3%; p_TOST is 1 for Jev and 1 for Haiku, and H4 is below alpha after Holm for neither. Both families track the gate's verdict on the lines the gate would have held and hold about half of the lines it allows (Cohen's kappa with the gate 0.470 and 0.423). Released on lines with a failed predicate, in anonymised EUR: people 4,759,520, Jev 0, Haiku 7,307, gate 0; for the gate, held lines go somewhere the log does not show. The expected correctives are the metric of registration section 6 (E4 row): each released line with a failed predicate carries the E2 O rate at W = 60 days of its defect group. The precedence and the unmatched-group rule are decision D-C-3: the H1 group if D1/D2 fails and otherwise the H2 group if D4 fails; a line failing neither (only D3, D6 or D7) has no E2 group and is counted, not imputed (8 sample lines, 1 of them released by people). The ERP view hands each model the

inputs of the predicates; this result is about the view as much as about the models. The expected-corrective totals are the number of corrective events the E2 rates would predict for the lines each arm released, under the stated assignment: a projection from observed rates onto the sample, not an observed count.

On the lines the people arm classifies. The same four arms, re-tabulated on the 1,848 sample lines the people arm classifies; the table above, on all sample lines, stays primary:

arm	held, lines with a failed predicate	held, clean lines	kappa with gate	value released on lines with a failed predicate	expected correctives
people (from the log)	301 of 912 (33.0%)	318 of 936 (34.0%)	n/a	4,759,520	12.260
Jev	912 of 912 (100.0%)	491 of 936 (52.5%)	0.472	0	0.000
Haiku	908 of 912 (99.6%)	524 of 936 (56.0%)	0.433	7,307	0.076
gate	912 of 912 (100.0%)	0 of 936 (0.0%)	n/a	0	0.000

Source: `results/e4/arms-classified-lines.json`; a re-tabulation on the lines the people arm classifies.

What the people arm measures. The log records no hold decision except a payment block. "Held by people" (section 3.6) is a payment block before the first clearing, or no clearing within W of the first invoice receipt: an inferred non-release, not the same act as a model's hold verdict at t*. Of the 53,262 framed lines held by people, 10 were held by a payment block alone, 53,182 by no clearing within W alone, and 70 by both:

reading	predicate status at t*	held by people	payment block before the first clearing	no clearing within W of the first invoice	both
literal (registered text)	defective	5,664	9	5,619	36
literal (registered text)	clean	1	1	0	0
literal (registered text)	held unknown	47,597	0	47,563	34
amended (D-C-2 B)	defective	5,664	9	5,619	36
amended (D-C-2 B)	clean	47,598	1	47,563	34
amended (D-C-2 B)	held unknown	0	0	0	0

Source: `results/e3/human-hold-composition.json`.

Exploratory: the event timeline instead of the ERP view. Labelled exploratory and outside the Holm family: the same lines, the same families and rules, with the line's events up to t* (date, activity, value) in place of the ERP view.

family (timeline view, exploratory)	held, lines with a failed predicate	held, clean lines	kappa with gate	invalid replies
Jev	126 of 1,000 (12.6%)	38 of 1,000 (3.8%)	0.088	0
Haiku	95 of 1,000 (9.5%)	39 of 1,000 (3.9%)	0.056	0

Source: `results/e4-exploratory/arms-exploratory.json`; exploratory, no interval, outside the Holm family.

Under the timeline view the same models held far fewer lines with a failed predicate (exploratory, no interval). Spend for both families and both views together was USD 1.57 against the registered cap of USD 60 (`results/e6/cost.json`).

5.5 Derivation (E5)

Paper 5's compiler, at the pinned commit and unmodified, was given per category the observed states of the evidence set and the three declared losses of the registration: L1, pay for goods not received (the category requires a goods receipt and none is uncancelled); L2, pay twice (the clearing's position among the line's clearings exceeds the invoices recorded); L3, pay a changed amount (a change since approval). The derived contract is the set of registered predicates whose observation is in the compiler's minimum-cardinality sufficient set. The comparison is the registered one (registration section 6, E5): "whether the derived contract equals the category contract in 3.1".

category	clearings	states	L1	L2	L3	fewest fields	derived contract	registered contract	result
2-way	188	3	0	0	4	`changes`	D4	D3, D4, D5, D6, D7	does not reproduce
3-way, invoice after receipt	9,640	896	0	81	1,585	`changes`, `clearing_ordinal`, `invoices`	D3, D4	D1, D2, D3, D4, D5, D6, D7	does not reproduce
3-way, invoice before receipt	145,706	758	692	1,390	15,568	`changes`, `clearing_ordinal`, `invoices`, `receipts`	D1, D2, D3, D4	D1, D2, D3, D4, D5, D6, D7	does not reproduce

The compiler does not reproduce the registered contract in any category. The three losses derive at most D1 to D4: no declared loss reads the observations of D5, D6 or D7, so those predicates correspond to no declared loss. In the before-receipt category the derived D1 to D4 is the three-way match in the narrow sense (receipt, invoice and an unchanged approved amount), and the registered contract carries three predicates beyond it. In the after-receipt category the compiler derives D3 and D4 only, because L1 occurs on 0 analysed clearings there, and the registered contract carries five predicates beyond it. This is a limit of deriving from one log: a loss the log never shows calls for no observation. Pay twice needs the clearing's position on its line, an observation no registered predicate reads.

Read the other way, paper 5's sufficiency function applied to the registered contract over the same states (`results/e5/sufficiency.json`):

category	registered fields	sufficient for	not sufficient for	clearings with L2	minimal	derived contract a subset
2-way	`invoices`, `changes`, `distinct_actors`, `blocks`, `closed`	L1, L2, L3	none	0	no	yes
3-way, invoice after receipt	`receipts`, `invoices`, `changes`, `distinct_actors`, `blocks`, `closed`	L1, L3	L2	81	no	yes
3-way, invoice before receipt	`receipts`, `invoices`, `changes`, `distinct_actors`, `blocks`, `closed`	L1, L3	L2	1,390	no	yes

The registered contract is sufficient, not derived: sufficient for the three declared losses in the two-way category, and in the three-way categories for L1 and L3 but not for L2, for the reason above; minimal in no category (C1). The compiler's contract is smaller in every category, and so C9, which asks whether it equals the registered one, is not supported.

5.6 Governance cost (E6)

The full replay of every line, from reading the log to the written evidence set, took a median of 153.5 seconds over three runs on a 4-core machine, 0.000987 seconds per gate decision. A model decision cost USD 0.000022 with Jev and USD 0.000368 with Haiku.

5.7 Replication on travel declarations (E7)

The unit is a payment on a declaration case at least W before the latest timestamp of its log; D1' approval present, D2' approval by the required role and D3' unchanged after approval are evaluated before the payment, and O' is a rejection or resubmission within W after it. Descriptive and secondary, outside the Holm family.

log	cases	payments	excluded after T_end' - W	units
domestic	10,500	10,044	2	10,042
international	6,449	6,187	0	6,187

scope	predicate	failing payments	O' rate, failing	O' rate, all-pass	difference	bootstrap 95%
domestic	D1'	2	0.500	0.000	0.500	[0.0000, 1.0000]
domestic	D2'	2	0.500	0.000	0.500	[0.0000, 1.0000]
domestic	D3'	0	n/a	0.000	n/a	n/a
international	D1'	0	n/a	0.000	n/a	n/a
international	D2'	0	n/a	0.000	n/a	n/a
international	D3'	0	n/a	0.000	n/a	n/a
pooled	D1'	2	0.500	0.000	0.500	[0.0000, 1.0000]
pooled	D2'	2	0.500	0.000	0.500	[0.0000, 1.0000]
pooled	D3'	0	n/a	0.000	n/a	n/a

The replication is inconclusive. Only 2 of 16,229 payments fail any predicate; the 2 domestic payments with D3' unknown are the same 2 that fail D1' and D2' (`results/e7/unknown-overlap.json`). O' follows 0 of the 16,227 payments with every predicate passing; no comparison has an interval excluding zero. A bootstrap replicate in which either group is empty is discarded and the percentiles are taken over the retained replicates (8,685 of 10,000 for the domestic log and 8,696 pooled, `results/e7/comparisons.json`; a deviation, Section 4). The registered names `Declaration FINAL_APPROVED by DIRECTOR` and `Declaration REJECTED by DIRECTOR` do not occur in the domestic log; they are reported with zero counts, not replaced.

6 Coverage

What the replay can and cannot see, as registered:

Detected (BPI 2019, E1 to E6)

defect class	predicate	dimension	detected from
goods receipt required but absent at clearing	D1 receipt_present	presence	receipts minus cancellations at most zero before t_c (count of `Record Goods Receipt` minus count of `Cancel Goods Receipt`), on a line whose category requires receipt
stale receipt: receipt recorded after clearing	D2 receipt_before_clearing	freshness, order	the first uncancelled `Record Goods Receipt` falls after t_c (lag band registered)
invoice absent at clearing	D3 invoice_present	presence	no `Record Invoice Receipt` before t_c

defect class	predicate	dimension	detected from
changed after approval	D4 unchanged_since_approval	validity	`Change Price` or `Change Quantity` between the last approval-bearing event and t_c
same actor created and cleared	D5 distinct_actors	authority	actor of `Create Purchase Order Item` equals actor of `Clear Invoice`
clearing under an open payment block	D6 block_respected	scope	count of `Set Payment Block` minus count of `Remove Payment Block` before t_c greater than zero
line deleted or blocked at clearing	D7 not_deleted	validity	count of `Delete Purchase Order Item` plus `Block Purchase Order Item` minus `Reactivate Purchase Order Item` before t_c greater than zero

Detected (BPI 2020 declarations, E7, secondary)

defect class	predicate	dimension	detected from
paid without an approval event	D1' approval_present	presence	no event in the registered approval list before the payment
approved by a role other than the required one	D2' approval_by_required_role	authority	the approval chain before payment lacks the registered required role
declaration changed after approval	D3' unchanged_since_approval	validity	a resubmission by the employee between the last approval and payment

Undetected, with reason

defect class	reason
wrong amounts in the payload	values are anonymised by a linear translation; the log carries cumulative net worth per event, not invoice line content, so a mismatch between invoice text and goods is not observable
fraud	intent is not an event attribute; no predicate over the log separates a fraudulent clearing from an erroneous one, and the paper makes no statement about it
vendor identity (wrong, duplicate or fictitious vendor)	vendor names and ids are anonymised consistently; the log carries no master-data change history or bank details
recovery of corrected payments	the log records the corrective event and nothing about recovery; whether money was returned, offset or written off is not recorded
what a held line becomes	a gate hold is a replay verdict; the log records the human path only, so the fate of a line the gate would have held is not observable
approvals outside the log	approvals given by e-mail, in another system or verbally leave no event; D4's "last approval-bearing event" sees only logged approvals
partial receipt and value mismatch per receipt	the replay checks presence and order of the receipt, not the matched values; values are anonymised and the registered mapping reads no per-receipt quantity
duplicate invoice across lines	the case is one purchase-order line and the registered mapping carries no invoice document id, so an invoice cleared on two lines is not linked

Per class, the failures found (BPI 2019 on analysed clearings, BPI 2020 on payments):

log	predicate	failed	unknown
BPI 2019 (analysed clearings)	D1	692	0
BPI 2019 (analysed clearings)	D2	692	0
BPI 2019 (analysed clearings)	D3	362	0
BPI 2019 (analysed clearings)	D4	17,157	0
BPI 2019 (analysed clearings)	D5	0	0
BPI 2019 (analysed clearings)	D6	4	0
BPI 2019 (analysed clearings)	D7	19	0
BPI 2020 domestic (payments)	D1'	2	0
BPI 2020 domestic (payments)	D2'	2	0
BPI 2020 domestic (payments)	D3'	0	2
BPI 2020 international (payments)	D1'	0	0
BPI 2020 international (payments)	D2'	0	0
BPI 2020 international (payments)	D3'	0	0

Sources: `results/e1/coverage-check.json`, `results/e1/predicate-unknowns.json` (BPI 2019 unknown), `results/e7/predicates.json`.

On BPI 2019 every detected class fails at least once except the authority class (D5), with 0 failed clearings: across the log, the 23 actors who record a `Clear Invoice` and the 185 who record a line's earliest creation share 0, so the two are separate populations; no `Clear Invoice` is recorded by a batch user (an actor whose name starts with the registered batch prefix), and 7,941 carry no actor (`results/e1/actors.json`), none of them on an analysed clearing (0; Section 5.2). D5 fires on synthetic lines in the engine's tests (`test_d5_actors` and `test_verdict_at_decision_point_d5_off_clearing` in `tests/test_engine_units.py`, and fixture line C in `tests/test_pipeline.py`). On BPI 2020, D3' fails on no payment in either log, and in the international log no predicate fails at all.

7 Limitations and non-claims

The outcome is a proxy: a corrective event after a clearing is not proof that the clearing was wrong, and the study is observational. Category, spend area, value, vendor activity and month are reported beside every estimate (Appendix D for exposure, Appendix E for H1, H2 and the gate by outcome) and never modelled in. Values are anonymised. What a line the gate would have held becomes is not in the log. The people column reports what the process did within W; the comparison across the people, the models and the gate is of outcomes of three deciders, not of three acts. The only inference the three columns support is this: on the 1,848 sample lines the log classifies, the process released about two thirds of both sets within W (people held 33.0% and 34.0%); the models shown the predicate inputs held nearly every line with a failed predicate (100.0% and 99.6%) and about half of the clean ones (52.5% and 56.0%; 53.0% and 57.3% on all 2,000 sample lines); the gate held exactly the lines with a failed predicate. No decision quality is inferred for people, whose column is a process outcome. The ERP view hands the model the inputs of the predicates; under the timeline view the same models held far fewer lines (exploratory, no interval), which is a statement about views, not about the models alone. The derivation depends on how the three declared losses are expressed over the evidence set; the expressions are recorded with the result. One company's process, one year.

8 Conclusion

11.41% of the 155,534 analysed invoice clearings in this process were made on evidence that failed at least one registered predicate, carrying 19.41% of the cleared value; those made with an absent or late receipt (D1/D2) were followed by a corrective at a higher rate than three-way clearings with every applicable predicate passing, and those made after a change since approval (D4 fail) at a higher rate than D4-pass clearings. The sufficiency check adds that the registered contract is sufficient for "goods not received" and "changed amount", and not for "pay twice" in the three-way categories, because no registered predicate reads the clearing's position on its line.

Acknowledgements

The prior-art fence was reviewed externally by Eduardo Arana after the tag `prereg-p7-v1` and before `prereg-p7-v1.1`, and he reviewed drafts v0.1 to v0.3. AI tools were used as support: builder sessions with Claude for the code, the runs and the drafting from slots; two machine prior-art sweeps, on ChatGPT and on Perplexity Computer (for the second, the platform made two statements, unverified in-chat self-reports of the platforms, in this order: a later session reported its own model as `Claude Opus 5.5` and stated that it produced the sweep, and the same session then identified the sweep's model as `DeepSeek V4 Pro`); and machine reviews of drafts v0.1 to v0.3, on ChatGPT and on Perplexity (the Perplexity reviews of v0.1 and v0.2 state their model as `Claude Opus 5.5`; that of v0.3 names no serving model). All are committed unedited. The author is solely responsible for every claim. We thank the organisers of the BPI Challenges and 4TU.ResearchData for the logs.

References

- Peyman Badakhshan, Samuel Gosling, Jerome Geyer-Klingeberg, Janina Nakladal, Johannes Schukat, Jennifer Gsenger (2019). Process Mining in the Coatings and Paints Industry: The Purchase Order Handling Process. BPI Challenge 2019 report, non-student category, ICPM 2019. https://icpmconference.org/2019/wp-content/uploads/sites/6/2019/07/BPI-Challenge-Submission-4.pdf
- Gaston Besanson (2026a). SARC: A Governance-by-Architecture Framework for Agentic AI Systems. arXiv:2605.07728 (paper 1 of the SARC series: the framework (may the agent act)). https://arxiv.org/abs/2605.07728
- Gaston Besanson (2026b). Green SARC: Predictive Cost and Carbon Governance for Agentic AI Systems. arXiv:2606.15954 (paper 2 of the SARC series: budget (cost and carbon)). https://arxiv.org/abs/2606.15954
- Gaston Besanson (2026c). SARC-DQ: Runtime Data-Quality Gating for Agentic AI: Silent Evidence Defects, the Incompetence Shield, and Downstream-Only Remediation. arXiv:2607.26313 (paper 3 of the SARC series: admit (data-quality gating)). https://arxiv.org/abs/2607.26313
- Gaston Besanson (2026d). One Gate Is Not Enough: Composing Stateful Pre-Action Controls for Agentic AI. arXiv:2608.18360 (paper 4 of the SARC series; repository sarc-suite-one-pass). https://arxiv.org/abs/2608.18360
- Gaston Besanson (2026e). Compiling Sufficient Governance Context from Declared Losses and Reachable States: Exact Observation-Contract Synthesis with Cardinality and Cost Objectives. arXiv:2609.26016 (paper 5 of this programme; CITATION.cff of sarc-authority-derivation). https://arxiv.org/pdf/2609.26016
- Gaston Besanson (2026f). Probabilistic Sensing, Deterministic Authority: Admitting Model-Produced Observations into Sufficiency-Checked Governance Contracts. arXiv:2610.10978 (paper 6 of the SARC series: sense (facts read from unstructured evidence)). https://arxiv.org/abs/2610.10978
- Kiarash Diba, Simon Remy, Luise Pufahl (2019). BPI Challenge 2019: Performance and Compliance Analysis of Procurement Processes Using Process Mining. BPI Challenge 2019 report, non-student category, ICPM 2019. https://icpmconference.org/2019/wp-content/uploads/sites/6/2019/07/BPI-Challenge-Submission-6.pdf
- Matthew Harding, Gabriel F. R. Vasconcelos (2022). Managers versus Machines: Do Algorithms Replicate Human Intuition in Credit Ratings?. arXiv:2202.04218 (submitted 2022-02-09). https://arxiv.org/abs/2202.04218
- ICPM 2019 (2019). BPI Challenge 2019 (ICPM 2019 conference page). https://icpmconference.org/2019/bpi-challenge/
- ICPM 2020 (2020a). Awards (ICPM 2020). https://icpmconference.org/2020/awards/

- ICPM 2020 (2020b). BPI Challenge 2020 (ICPM 2020 conference page). https://icpmconference.org/2020/bpi-challenge/
- Md Abdul Kadir, Sai Suresh Macharla Vasu, Sidharth S. Nair, Daniel Sonntag (2025). AuditCopilot: Leveraging LLMs for Fraud Detection in Double-Entry Bookkeeping. arXiv:2512.02726 (submitted 2025-12-02). https://arxiv.org/abs/2512.02726
- Stijn Kas, Ruben Post, Sebastiaan Wiewel (2020). Automated Machine Learning in a Process Mining Context. BPI Student Challenge Award, ICPM 2020. https://icpmconference.org/2020/wp-content/uploads/sites/4/2020/10/ICPM_2020_paper_44.pdf
- Linh Thao Ly, Fabrizio Maria Maggi, Marco Montali, Stefanie Rinderle-Ma, Wil M.P. van der Aalst (2015). Compliance monitoring in business processes: Functionalities, application, and tool-support. Information Systems 54, 209-234 (December 2015), Elsevier. doi:10.1016/j.is.2015.02.007
- Vincent Finn Alexander Meyer zu Wickern, Mounisha Juluru, Anshu Roy, Viet-Hung Vu, Thi-Thu-Hang Nguyen (2019). Analysis and prediction of purchasing compliance using process mining. BPI Challenge 2019 report, student category, ICPM 2019. https://icpmconference.org/2019/wp-content/uploads/sites/6/2019/07/BPI-Challenge-Student-Submission-4.pdf
- Hoang Nguyen, Marlon Dumas, Marcello La Rosa, Fabrizio Maria Maggi, Suriadi Suriadi (2016). Business Process Deviance Mining: Review and Evaluation. arXiv:1608.08252 (submitted 2016-08-29). https://arxiv.org/abs/1608.08252
- Nesi Outmazgin, Pnina Soffer (2013). Business Process Workarounds: What Can and Cannot Be Detected by Process Mining. Enterprise, Business-Process and Information Systems Modeling (BPMDS 2013, EMMSAD 2013), LNBIP vol. 147, pp. 48-62, Springer. doi:10.1007/978-3-642-38484-4_5
- Diego Palma, Kyu Bin Kim, Zhen Han, Allbright Dsouza, Zhiyuan Liu (2026). Business Compromise Detection with Agentic AI and LLM-driven Knowledge Discovery. arXiv:2609.32643 (submitted 2026-09-26). https://arxiv.org/html/2609.32643
- Public Company Accounting Oversight Board (2004). An Audit of Internal Control Over Financial Reporting Performed in Conjunction with an Audit of Financial Statements (PCAOB Release No. 2004-001). PCAOB Release 2004-001, March 9, 2004. https://pcaobus.org/Rulemaking/Docket008/2004-03-09_Release_2004-001-all.pdf
- Mor Peleg, Pnina Soffer, Johny Ghattas (2008). Mining Process Execution and Outcomes – Position Paper. Business Process Management Workshops (BPM 2007), LNCS vol. 4928, pp. 395-400, Springer. doi:10.1007/978-3-540-78238-4_40
- Maja Pesic, Helen Schonenberg, Wil M.P. van der Aalst (2007). DECLARE: Full Support for Loosely-Structured Processes. 11th IEEE International Enterprise Distributed Object Computing Conference (EDOC 2007), p. 287, IEEE. doi:10.1109/EDOC.2007.14
- Adrien Porter, David Masse, Nuss Visatemongkolchai, Jithendra Seneviratne, Tanvee Deokule, Nicholas Hartman (2019). Balancing Efficiency and Risk in Procure to Pay: Safely Realizing Cost Savings Using Process Mining Techniques. BPI Challenge 2019 report, non-student category, ICPM 2019. https://icpmconference.org/2019/wp-content/uploads/sites/6/2019/07/BPI-Challenge-Submission-8.pdf
- Luise Pufahl, Richard Hobeck, Paul Binetruy, Wepan Chada, Mykola Digtiar, Kim Julian Gülle, Marta Slarzynska, Fabian Stiehle, Ingo Weber (2020). Performance, Variant, and Conformance Analysis of an Academic Travel Reimbursement Process. BPI Challenge Award, ICPM 2020. https://icpmconference.org/2020/wp-content/uploads/sites/4/2020/10/ICPM_2020_paper_148.pdf
- Irene Teinemaa, Marlon Dumas, Marcello La Rosa, Fabrizio Maria Maggi (2019). Outcome-Oriented Predictive Process Monitoring. ACM Transactions on Knowledge Discovery from Data 13(2), 2019. doi:10.1145/3301300
- Wil van der Aalst (2016). Process Mining: Data Science in Action. Springer Berlin, Heidelberg, second edition. doi:10.1007/978-3-662-49851-4

- Boudewijn van Dongen. BPI Challenge 2019. 4TU.ResearchData, CC BY 4.0. doi:10.4121/uuid:d06aff4b-79f0-45e6-8ec8-e19730c248f1
- Boudewijn van Dongen. BPI Challenge 2020: International Declarations. 4TU.ResearchData, CC BY-NC 4.0. doi:10.4121/uuid:2bbf8f6a-fc50-48eb-aa9e-c4ea5ef7e8c5
- Boudewijn van Dongen. BPI Challenge 2020: Domestic Declarations. 4TU.ResearchData, CC BY-NC 4.0. doi:10.4121/uuid:3f422315-ed9d-4882-891f-e180b5b4feb5

Appendix A: Registration summary

The registered numbers, Section 9 of the registration:

item	value
window	Create Purchase Order Item in calendar 2018 (UTC)
W	60 days; sensitivity 30 days
lag bands	[0, 1], (1, 7], (7, 30], over 30 days, never
K (framework exclusion)	more than 12 clearings
E0	500 lines
E4	N_d = 1,000, N_c = 1,000; m = 0.05
alpha	0.05, Holm over H1, H2, H3, H4-Jev, H4-Haiku
bootstrap	B = 10,000, purchase-document clusters
seed	20261201
cap	USD 60
pins	`engines.lock` at this commit

Appendix B: Claims, decisions and deviations

Claim statuses, each set from its slot file:

claim	status	evidence
C1	supported	`results/e5/sufficiency.json`, `results/e1/evidence-set.jsonl.gz`, `tests/test_properties.py`; engine evidence `results/mutation.json`
C2	supported	`results/e1/exposure.json`, `results/e1/exposure-by-stratum.json`
C3	supported	`results/e2/hypotheses.json`
C4	supported	`results/e2/hypotheses.json`
C5	inconclusive: defect class has zero incidence in the analysed clearings	`results/e1/coverage-check.json`, `results/e2/hypotheses.json`
C6	not supported for both families	`results/e4/h4.json`
C7	supported	`results/e3/gate-held.json`, `results/e3/verdict-by-outcome.json`, `results/e3/lag-bands.json`
C8	supported	`results/e4/arms.json`
C9	not supported: 2-way does not reproduce; 3-way, invoice after receipt does not reproduce; 3-way, invoice before receipt does not reproduce	`results/e5/derivation.json`
C10	supported	`results/e6/timing-e1.json`, `results/e6/cost.json`
C11	inconclusive	`results/e7/comparisons.json`
C12	supported; D5 class zero incidence	`results/e1/coverage-check.json`
C13	supported for the internal double runs; external attempt 1 (issue #1): every reported number regenerated, numeric slots equal within tolerance, run incomplete on a tooling defect corrected in 1.0.1; re-run pending	`results/reproducibility.json`, `results/e4/double-run.json`, `results/reproduction/attempt-1.json`

Reading notes, as in the claims file. C1's "(paper 5)" refers to paper 5's contract construct, whose sufficiency for the declared losses over the observed states E5 checks with paper 5's own function, per category and per loss (`results/e5/sufficiency.json`): sufficient for every declared loss in the two-way category, and for L1 and L3 but not L2 in the three-way categories; minimal in none; each predicate checks one SARC-DQ dimension (Section 3). C2's "relied on" is read as "made on evidence that failed". C9's "per-category matching rule" is the category contract of section 3.1, the comparator the registration's E5 row fixes; the before-receipt category recovers the narrow three-way match (D1 to D4), and the status "not supported" refers to the registered comparison (Section 5.5).

Every decision taken while the work ran is in the repository's decision log, from D-A0-1 to D-D-4, and every deviation from the tagged registration is in its deviations file (Section 4).

Appendix C: Reproduction

`make bootstrap` clones the sibling repositories at their pins, creates the environment, fetches the logs and checks their hashes; `make test` and `make lint` run the tests and the lints; `make gates` runs the release gates G1 to G9; the experiment runners regenerate every slot under `results/`, and `make paper` regenerates this paper and its figures from the slots. Internal double runs on one machine are byte-identical (16 files for E1 to E3). Across machines and architectures (x86-64 Linux at the builder; arm64 macOS in the first external attempt, issue #1), every result file is equal to within 1e-12 relative; the last floating-point digits can differ (maximum observed absolute difference 1.1e-16); figure bytes differ by font embedding; the paper text, its TeX source and the README regenerate byte-identically. External reproduction: attempt 1 on v1.0.0 did not complete because of a tooling error in the optional PDF step, corrected in 1.0.1; the re-run is pending.

Appendix D: Exposure by stratum

The share of analysed clearings and of cleared value per defect class (a failed predicate), per stratum, with the purchase-document-clustered 95% intervals; classes with no clearing in a stratum are omitted here and kept in `results/e1/exposure-by-stratum.json`. Company and vendor strata are in the slot file only, as hashed labels. The spend-area labels are the public log's own strings and the registered strata; the challenge itself names the sector.

Item category

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
2-way	D4	4 of 188	2.13%	[0.00%, 4.94%]	2.68%	[0.00%, 8.71%]
2-way	at least one defect	4 of 188	2.13%	[0.00%, 4.94%]	2.68%	[0.00%, 8.71%]
3-way, invoice after receipt	D3	9 of 9,640	0.09%	[0.01%, 0.21%]	0.02%	[0.00%, 0.05%]
3-way, invoice after receipt	D4	1,585 of 9,640	16.44%	[14.36%, 18.73%]	15.65%	[9.48%, 24.64%]
3-way, invoice after receipt	at least one defect	1,592 of 9,640	16.51%	[14.42%, 18.79%]	15.67%	[9.50%, 24.66%]
3-way, invoice before receipt	D1 class (absent receipt)	610 of 145,706	0.42%	[0.32%, 0.53%]	0.21%	[0.14%, 0.28%]
3-way, invoice before receipt	D2 class (stale receipt)	82 of 145,706	0.06%	[0.02%, 0.10%]	0.22%	[0.05%, 0.45%]
3-way, invoice before receipt	D3	353 of 145,706	0.24%	[0.15%, 0.35%]	0.04%	[0.02%, 0.06%]
3-way, invoice before receipt	D4	15,568 of 145,706	10.68%	[10.21%, 11.19%]	21.92%	[20.71%, 23.18%]
3-way, invoice before receipt	D6	4 of 145,706	0.00%	[0.00%, 0.01%]	0.05%	[0.00%, 0.16%]
3-way, invoice before receipt	D7	19 of 145,706	0.01%	[0.01%, 0.02%]	0.02%	[0.00%, 0.04%]

3-way, invoice before receipt	at least one defect	16,152 of 145,706	11.09%	[10.60%, 11.60%]	22.21%	[21.00%, 23.48%]

Spend area

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
Additives	D1 class (absent receipt)	9 of 10,206	0.09%	[0.04%, 0.15%]	0.13%	[0.02%, 0.29%]
Additives	D2 class (stale receipt)	4 of 10,206	0.04%	[0.01%, 0.08%]	0.02%	[0.00%, 0.04%]
Additives	D3	4 of 10,206	0.04%	[0.01%, 0.08%]	0.01%	[0.00%, 0.04%]
Additives	D4	1,207 of 10,206	11.83%	[11.10%, 12.57%]	13.94%	[12.60%, 15.32%]
Additives	D7	1 of 10,206	0.01%	[0.00%, 0.03%]	0.01%	[0.00%, 0.02%]
Additives	at least one defect	1,219 of 10,206	11.94%	[11.22%, 12.69%]	14.07%	[12.74%, 15.46%]
CAPEX & SOCS	D1 class (absent receipt)	46 of 4,672	0.98%	[0.62%, 1.40%]	0.74%	[0.38%, 1.25%]
CAPEX & SOCS	D2 class (stale receipt)	9 of 4,672	0.19%	[0.00%, 0.50%]	0.10%	[0.00%, 0.31%]
CAPEX & SOCS	D3	15 of 4,672	0.32%	[0.11%, 0.59%]	0.06%	[0.01%, 0.13%]
CAPEX & SOCS	D4	278 of 4,672	5.95%	[4.50%, 7.90%]	25.55%	[12.52%, 38.45%]
CAPEX & SOCS	at least one defect	332 of 4,672	7.11%	[5.57%, 9.12%]	26.35%	[13.48%, 39.09%]
Chemicals & Intermediates	at least one defect	0 of 1	0.00%	[0.00%, 0.00%]	0.00%	[0.00%, 0.00%]
Commodity Resins	D1 class (absent receipt)	1 of 306	0.33%	[0.00%, 1.10%]	0.79%	[0.00%, 2.69%]
Commodity Resins	D4	80 of 306	26.14%	[19.93%, 32.47%]	27.89%	[19.61%, 36.60%]
Commodity Resins	at least one defect	81 of 306	26.47%	[20.18%, 32.78%]	28.67%	[20.42%, 37.47%]
Energy	at least one defect	0 of 3	0.00%	[0.00%, 0.00%]	0.00%	[0.00%, 0.00%]
Enterprise Services	D1 class (absent receipt)	1 of 370	0.27%	[0.00%, 0.90%]	0.09%	[0.00%, 0.39%]
Enterprise Services	D2 class (stale receipt)	1 of 370	0.27%	[0.00%, 0.93%]	0.01%	[0.00%, 0.03%]
Enterprise Services	D4	20 of 370	5.41%	[2.55%, 9.09%]	7.21%	[1.89%, 18.58%]
Enterprise Services	at least one defect	22 of 370	5.95%	[2.98%, 9.83%]	7.30%	[1.95%, 18.80%]
Latex & Monomers	D1 class (absent receipt)	7 of 2,584	0.27%	[0.07%, 0.54%]	0.26%	[0.05%, 0.55%]
Latex & Monomers	D2 class (stale receipt)	3 of 2,584	0.12%	[0.00%, 0.27%]	0.15%	[0.00%, 0.36%]
Latex & Monomers	D3	1 of 2,584	0.04%	[0.00%, 0.12%]	0.01%	[0.00%, 0.02%]
Latex & Monomers	D4	479 of 2,584	18.54%	[16.86%, 20.28%]	18.46%	[16.67%, 20.40%]
Latex & Monomers	D7	1 of 2,584	0.04%	[0.00%, 0.12%]	0.05%	[0.00%, 0.16%]
Latex & Monomers	at least one defect	488 of 2,584	18.89%	[17.18%, 20.63%]	18.82%	[17.02%, 20.76%]
Logistics	D3	1 of 2,069	0.05%	[0.00%, 0.16%]	0.01%	[0.00%, 0.03%]
Logistics	D4	114 of 2,069	5.51%	[3.09%, 8.13%]	6.11%	[1.88%, 15.39%]

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
Logistics	at least one defect	114 of 2,069	5.51%	[3.09%, 8.13%]	6.11%	[1.88%, 15.39%]
Marketing	D1 class (absent receipt)	6 of 1,220	0.49%	[0.00%, 1.31%]	0.03%	[0.00%, 0.12%]
Marketing	D4	108 of 1,220	8.85%	[5.94%, 12.34%]	47.19%	[25.71%, 59.53%]
Marketing	at least one defect	114 of 1,220	9.34%	[6.36%, 12.92%]	47.22%	[25.76%, 59.54%]
Others	D1 class (absent receipt)	8 of 1,776	0.45%	[0.00%, 1.24%]	0.00%	[0.00%, 0.00%]
Others	D4	86 of 1,776	4.84%	[3.52%, 6.31%]	9.16%	[2.55%, 22.87%]
Others	at least one defect	94 of 1,776	5.29%	[3.85%, 6.89%]	9.16%	[2.55%, 22.88%]
Packaging	D1 class (absent receipt)	136 of 63,561	0.21%	[0.16%, 0.27%]	0.24%	[0.12%, 0.39%]
Packaging	D2 class (stale receipt)	34 of 63,561	0.05%	[0.00%, 0.15%]	0.04%	[0.00%, 0.09%]
Packaging	D3	15 of 63,561	0.02%	[0.01%, 0.04%]	0.02%	[0.01%, 0.05%]
Packaging	D4	6,901 of 63,561	10.86%	[10.31%, 11.45%]	18.95%	[17.63%, 20.33%]
Packaging	D7	4 of 63,561	0.01%	[0.00%, 0.01%]	0.00%	[0.00%, 0.01%]
Packaging	at least one defect	7,035 of 63,561	11.07%	[10.52%, 11.66%]	19.13%	[17.82%, 20.53%]
Pigments & Colorants	D1 class (absent receipt)	3 of 1,694	0.18%	[0.00%, 0.41%]	0.24%	[0.00%, 0.67%]
Pigments & Colorants	D2 class (stale receipt)	2 of 1,694	0.12%	[0.00%, 0.37%]	0.19%	[0.00%, 0.60%]
Pigments & Colorants	D3	1 of 1,694	0.06%	[0.00%, 0.19%]	0.17%	[0.00%, 0.56%]
Pigments & Colorants	D4	295 of 1,694	17.41%	[15.35%, 19.59%]	29.19%	[23.93%, 34.78%]
Pigments & Colorants	at least one defect	297 of 1,694	17.53%	[15.48%, 19.71%]	29.37%	[24.09%, 34.95%]
Real Estate	D4	5 of 151	3.31%	[0.60%, 7.25%]	12.38%	[0.45%, 25.30%]
Real Estate	at least one defect	5 of 151	3.31%	[0.60%, 7.25%]	12.38%	[0.45%, 25.30%]
Sales	D1 class (absent receipt)	126 of 46,891	0.27%	[0.17%, 0.41%]	0.32%	[0.12%, 0.59%]
Sales	D2 class (stale receipt)	14 of 46,891	0.03%	[0.00%, 0.08%]	0.02%	[0.00%, 0.05%]
Sales	D3	62 of 46,891	0.13%	[0.05%, 0.26%]	0.10%	[0.04%, 0.19%]
Sales	D4	3,244 of 46,891	6.92%	[6.34%, 7.55%]	15.91%	[13.78%, 18.18%]
Sales	D7	10 of 46,891	0.02%	[0.00%, 0.05%]	0.05%	[0.00%, 0.14%]
Sales	at least one defect	3,378 of 46,891	7.20%	[6.61%, 7.84%]	16.24%	[14.14%, 18.48%]
Solvents	D3	3 of 1,573	0.19%	[0.00%, 0.43%]	0.04%	[0.00%, 0.09%]
Solvents	D4	216 of 1,573	13.73%	[11.80%, 15.69%]	16.61%	[13.63%, 19.78%]
Solvents	at least one defect	219 of 1,573	13.92%	[12.00%, 15.87%]	16.64%	[13.67%, 19.81%]
Specialty Resins	D4	228 of 1,295	17.61%	[15.33%, 19.94%]	17.00%	[14.21%, 19.98%]
Specialty Resins	at least one defect	228 of 1,295	17.61%	[15.33%, 19.94%]	17.00%	[14.21%, 19.98%]
Spend Area Unidentified	D4	3 of 18	16.67%	[0.00%, 40.00%]	9.65%	[0.00%, 30.06%]

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
Spend Area Unidentified	at least one defect	3 of 18	16.67%	[0.00%, 40.00%]	9.65%	[0.00%, 30.06%]
Titanium Dioxides	D1 class (absent receipt)	3 of 429	0.70%	[0.00%, 1.59%]	0.10%	[0.00%, 0.27%]
Titanium Dioxides	D2 class (stale receipt)	2 of 429	0.47%	[0.00%, 1.21%]	1.66%	[0.00%, 4.28%]
Titanium Dioxides	D4	119 of 429	27.74%	[22.84%, 32.89%]	36.36%	[28.19%, 44.61%]
Titanium Dioxides	D6	2 of 429	0.47%	[0.00%, 1.51%]	0.59%	[0.00%, 1.98%]
Titanium Dioxides	at least one defect	124 of 429	28.90%	[23.95%, 34.11%]	37.05%	[28.83%, 45.32%]
Trading & End Products	D1 class (absent receipt)	261 of 15,104	1.73%	[0.93%, 2.68%]	0.07%	[0.03%, 0.13%]
Trading & End Products	D2 class (stale receipt)	12 of 15,104	0.08%	[0.01%, 0.16%]	0.13%	[0.02%, 0.28%]
Trading & End Products	D3	259 of 15,104	1.71%	[0.93%, 2.68%]	0.17%	[0.05%, 0.32%]
Trading & End Products	D4	3,339 of 15,104	22.11%	[18.90%, 25.57%]	43.14%	[38.19%, 47.92%]
Trading & End Products	D7	3 of 15,104	0.02%	[0.00%, 0.05%]	0.01%	[0.00%, 0.04%]
Trading & End Products	at least one defect	3,556 of 15,104	23.54%	[20.24%, 27.05%]	43.47%	[38.53%, 48.25%]
Workforce Services	D4	62 of 281	22.06%	[15.93%, 28.45%]	28.17%	[19.43%, 36.45%]
Workforce Services	at least one defect	62 of 281	22.06%	[15.93%, 28.45%]	28.17%	[19.43%, 36.45%]
missing	D1 class (absent receipt)	3 of 1,330	0.23%	[0.00%, 0.51%]	0.17%	[0.00%, 0.44%]
missing	D2 class (stale receipt)	1 of 1,330	0.08%	[0.00%, 0.24%]	0.33%	[0.00%, 1.04%]
missing	D3	1 of 1,330	0.08%	[0.00%, 0.24%]	0.01%	[0.00%, 0.03%]
missing	D4	373 of 1,330	28.05%	[25.10%, 30.94%]	28.16%	[23.41%, 32.95%]
missing	D6	2 of 1,330	0.15%	[0.00%, 0.47%]	0.04%	[0.00%, 0.12%]
missing	at least one defect	377 of 1,330	28.35%	[25.34%, 31.29%]	28.54%	[23.77%, 33.37%]

Month of the clearing

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
2018-01	D1 class (absent receipt)	1 of 255	0.39%	[0.00%, 1.68%]	3.62%	[0.00%, 12.12%]
2018-01	D3	1 of 255	0.39%	[0.00%, 1.68%]	3.62%	[0.00%, 12.12%]
2018-01	D4	8 of 255	3.14%	[0.73%, 7.95%]	9.26%	[0.10%, 22.02%]
2018-01	at least one defect	9 of 255	3.53%	[0.94%, 8.63%]	12.88%	[0.34%, 26.75%]
2018-02	D1 class (absent receipt)	5 of 2,812	0.18%	[0.00%, 0.41%]	0.01%	[0.00%, 0.02%]
2018-02	D2 class (stale receipt)	1 of 2,812	0.04%	[0.00%, 0.12%]	0.34%	[0.00%, 1.31%]
2018-02	D3	2 of 2,812	0.07%	[0.00%, 0.19%]	0.03%	[0.00%, 0.12%]

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
2018-02	D4	96 of 2,812	3.41%	[2.60%, 4.32%]	2.37%	[1.21%, 4.41%]
2018-02	at least one defect	103 of 2,812	3.66%	[2.83%, 4.59%]	2.75%	[1.38%, 5.13%]
2018-03	D1 class (absent receipt)	19 of 9,790	0.19%	[0.07%, 0.36%]	0.06%	[0.01%, 0.12%]
2018-03	D2 class (stale receipt)	4 of 9,790	0.04%	[0.00%, 0.10%]	0.10%	[0.00%, 0.29%]
2018-03	D3	11 of 9,790	0.11%	[0.02%, 0.26%]	0.04%	[0.00%, 0.11%]
2018-03	D4	855 of 9,790	8.73%	[7.37%, 10.37%]	8.36%	[6.02%, 11.75%]
2018-03	D7	2 of 9,790	0.02%	[0.00%, 0.05%]	0.00%	[0.00%, 0.02%]
2018-03	at least one defect	877 of 9,790	8.96%	[7.58%, 10.61%]	8.51%	[6.14%, 11.95%]
2018-04	D1 class (absent receipt)	87 of 15,063	0.58%	[0.21%, 1.06%]	0.15%	[0.01%, 0.36%]
2018-04	D2 class (stale receipt)	6 of 15,063	0.04%	[0.01%, 0.07%]	0.02%	[0.00%, 0.07%]
2018-04	D3	73 of 15,063	0.48%	[0.13%, 0.97%]	0.01%	[0.00%, 0.02%]
2018-04	D4	1,737 of 15,063	11.53%	[10.29%, 12.95%]	15.08%	[12.24%, 18.36%]
2018-04	D6	2 of 15,063	0.01%	[0.00%, 0.04%]	0.32%	[0.00%, 1.06%]
2018-04	D7	2 of 15,063	0.01%	[0.00%, 0.04%]	0.01%	[0.00%, 0.03%]
2018-04	at least one defect	1,827 of 15,063	12.13%	[10.83%, 13.61%]	15.54%	[12.63%, 18.94%]
2018-05	D1 class (absent receipt)	60 of 18,939	0.32%	[0.15%, 0.58%]	0.09%	[0.03%, 0.19%]
2018-05	D2 class (stale receipt)	8 of 18,939	0.04%	[0.01%, 0.10%]	0.03%	[0.00%, 0.07%]
2018-05	D3	38 of 18,939	0.20%	[0.05%, 0.46%]	0.03%	[0.01%, 0.06%]
2018-05	D4	1,909 of 18,939	10.08%	[9.27%, 10.95%]	16.92%	[10.76%, 25.85%]
2018-05	D7	1 of 18,939	0.01%	[0.00%, 0.02%]	0.00%	[0.00%, 0.01%]
2018-05	at least one defect	1,972 of 18,939	10.41%	[9.57%, 11.30%]	17.03%	[10.85%, 25.97%]
2018-06	D1 class (absent receipt)	70 of 19,748	0.35%	[0.20%, 0.53%]	0.10%	[0.03%, 0.21%]
2018-06	D2 class (stale receipt)	9 of 19,748	0.05%	[0.01%, 0.09%]	0.10%	[0.01%, 0.25%]
2018-06	D3	44 of 19,748	0.22%	[0.08%, 0.39%]	0.01%	[0.00%, 0.01%]
2018-06	D4	2,261 of 19,748	11.45%	[10.30%, 12.69%]	15.09%	[11.54%, 19.60%]
2018-06	at least one defect	2,331 of 19,748	11.80%	[10.64%, 13.06%]	15.28%	[11.68%, 19.82%]
2018-07	D1 class (absent receipt)	74 of 20,019	0.37%	[0.20%, 0.61%]	0.07%	[0.02%, 0.17%]
2018-07	D2 class (stale receipt)	1 of 20,019	0.00%	[0.00%, 0.02%]	0.00%	[0.00%, 0.00%]
2018-07	D3	32 of 20,019	0.16%	[0.03%, 0.37%]	0.01%	[0.00%, 0.01%]
2018-07	D4	1,991 of 20,019	9.95%	[8.93%, 11.07%]	17.30%	[13.42%, 21.79%]
2018-07	D6	2 of 20,019	0.01%	[0.00%, 0.03%]	0.00%	[0.00%, 0.01%]
2018-07	D7	1 of 20,019	0.00%	[0.00%, 0.02%]	0.00%	[0.00%, 0.01%]
2018-07	at least one defect	2,060 of 20,019	10.29%	[9.26%, 11.43%]	17.38%	[13.49%, 21.90%]
2018-08	D1 class (absent receipt)	56 of 21,610	0.26%	[0.18%, 0.35%]	0.13%	[0.06%, 0.23%]
2018-08	D2 class (stale receipt)	40 of 21,610	0.19%	[0.02%, 0.48%]	0.07%	[0.01%, 0.15%]
2018-08	D3	13 of 21,610	0.06%	[0.02%, 0.11%]	0.03%	[0.00%, 0.08%]

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
2018-08	D4	2,442 of 21,610	11.30%	[10.18%, 12.58%]	22.47%	[17.18%, 28.35%]
2018-08	at least one defect	2,531 of 21,610	11.71%	[10.55%, 13.02%]	22.66%	[17.38%, 28.56%]
2018-09	D1 class (absent receipt)	77 of 19,002	0.41%	[0.26%, 0.58%]	0.33%	[0.15%, 0.56%]
2018-09	D2 class (stale receipt)	1 of 19,002	0.01%	[0.00%, 0.02%]	0.00%	[0.00%, 0.00%]
2018-09	D3	23 of 19,002	0.12%	[0.03%, 0.23%]	0.06%	[0.01%, 0.14%]
2018-09	D4	2,253 of 19,002	11.86%	[10.83%, 12.97%]	27.43%	[23.69%, 31.34%]
2018-09	D7	5 of 19,002	0.03%	[0.01%, 0.06%]	0.05%	[0.00%, 0.16%]
2018-09	at least one defect	2,314 of 19,002	12.18%	[11.14%, 13.29%]	27.67%	[23.93%, 31.56%]
2018-10	D1 class (absent receipt)	91 of 19,193	0.47%	[0.15%, 0.96%]	0.06%	[0.02%, 0.11%]
2018-10	D2 class (stale receipt)	9 of 19,193	0.05%	[0.00%, 0.15%]	0.00%	[0.00%, 0.01%]
2018-10	D3	64 of 19,193	0.33%	[0.03%, 0.82%]	0.03%	[0.01%, 0.07%]
2018-10	D4	2,387 of 19,193	12.44%	[10.93%, 14.41%]	26.80%	[21.62%, 32.61%]
2018-10	D7	4 of 19,193	0.02%	[0.00%, 0.05%]	0.01%	[0.00%, 0.04%]
2018-10	at least one defect	2,483 of 19,193	12.94%	[11.37%, 14.95%]	26.87%	[21.69%, 32.69%]
2018-11	D1 class (absent receipt)	70 of 9,103	0.77%	[0.15%, 1.70%]	0.02%	[0.01%, 0.05%]
2018-11	D2 class (stale receipt)	3 of 9,103	0.03%	[0.00%, 0.08%]	1.29%	[0.00%, 3.45%]
2018-11	D3	61 of 9,103	0.67%	[0.05%, 1.60%]	0.04%	[0.00%, 0.10%]
2018-11	D4	1,218 of 9,103	13.38%	[11.90%, 15.02%]	22.30%	[15.43%, 30.79%]
2018-11	D7	4 of 9,103	0.04%	[0.00%, 0.10%]	0.01%	[0.00%, 0.05%]
2018-11	at least one defect	1,241 of 9,103	13.63%	[12.14%, 15.29%]	22.36%	[15.48%, 30.86%]

Line value quartile

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
Q1	D1 class (absent receipt)	369 of 38,549	0.96%	[0.63%, 1.34%]	0.59%	[0.42%, 0.79%]
Q1	D2 class (stale receipt)	17 of 38,549	0.04%	[0.01%, 0.09%]	0.05%	[0.01%, 0.12%]
Q1	D3	273 of 38,549	0.71%	[0.40%, 1.08%]	0.32%	[0.18%, 0.51%]
Q1	D4	2,012 of 38,549	5.22%	[4.42%, 6.30%]	5.01%	[4.45%, 5.64%]
Q1	D7	3 of 38,549	0.01%	[0.00%, 0.02%]	0.01%	[0.00%, 0.03%]
Q1	at least one defect	2,334 of 38,549	6.05%	[5.20%, 7.15%]	5.60%	[5.01%, 6.26%]
Q2	D1 class (absent receipt)	91 of 38,481	0.24%	[0.17%, 0.32%]	0.23%	[0.16%, 0.31%]
Q2	D2 class (stale receipt)	18 of 38,481	0.05%	[0.01%, 0.11%]	0.04%	[0.01%, 0.09%]
Q2	D3	35 of 38,481	0.09%	[0.04%, 0.16%]	0.09%	[0.04%, 0.16%]
Q2	D4	2,629 of 38,481	6.83%	[6.40%, 7.28%]	7.01%	[6.57%, 7.46%]
Q2	D7	3 of 38,481	0.01%	[0.00%, 0.02%]	0.01%	[0.00%, 0.02%]

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
Q2	at least one defect	2,724 of 38,481	7.08%	[6.64%, 7.54%]	7.24%	[6.79%, 7.71%]
Q3	D1 class (absent receipt)	67 of 38,963	0.17%	[0.12%, 0.23%]	0.17%	[0.12%, 0.23%]
Q3	D2 class (stale receipt)	20 of 38,963	0.05%	[0.01%, 0.11%]	0.04%	[0.01%, 0.08%]
Q3	D3	29 of 38,963	0.07%	[0.04%, 0.12%]	0.07%	[0.04%, 0.12%]
Q3	D4	4,447 of 38,963	11.41%	[10.77%, 12.11%]	12.35%	[11.60%, 13.17%]
Q3	D7	3 of 38,963	0.01%	[0.00%, 0.02%]	0.01%	[0.00%, 0.02%]
Q3	at least one defect	4,522 of 38,963	11.61%	[10.96%, 12.31%]	12.53%	[11.78%, 13.34%]
Q4	D1 class (absent receipt)	83 of 39,541	0.21%	[0.15%, 0.27%]	0.11%	[0.07%, 0.16%]
Q4	D2 class (stale receipt)	27 of 39,541	0.07%	[0.04%, 0.10%]	0.13%	[0.03%, 0.29%]
Q4	D3	25 of 39,541	0.06%	[0.03%, 0.10%]	0.02%	[0.01%, 0.04%]
Q4	D4	8,069 of 39,541	20.41%	[19.34%, 21.54%]	19.90%	[16.24%, 23.97%]
Q4	D6	4 of 39,541	0.01%	[0.00%, 0.03%]	0.03%	[0.00%, 0.11%]
Q4	D7	10 of 39,541	0.03%	[0.01%, 0.05%]	0.01%	[0.00%, 0.03%]
Q4	at least one defect	8,168 of 39,541	20.66%	[19.60%, 21.78%]	20.07%	[16.39%, 24.17%]

Vendor activity quartile

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
Q1	D1 class (absent receipt)	114 of 38,820	0.29%	[0.22%, 0.38%]	0.09%	[0.05%, 0.14%]
Q1	D2 class (stale receipt)	26 of 38,820	0.07%	[0.03%, 0.11%]	0.14%	[0.01%, 0.34%]
Q1	D3	45 of 38,820	0.12%	[0.06%, 0.18%]	0.03%	[0.01%, 0.05%]
Q1	D4	6,056 of 38,820	15.60%	[14.97%, 16.25%]	18.93%	[14.45%, 24.29%]
Q1	D6	4 of 38,820	0.01%	[0.00%, 0.03%]	0.04%	[0.00%, 0.15%]
Q1	D7	10 of 38,820	0.03%	[0.01%, 0.06%]	0.00%	[0.00%, 0.01%]
Q1	at least one defect	6,194 of 38,820	15.96%	[15.31%, 16.62%]	19.08%	[14.58%, 24.46%]
Q2	D1 class (absent receipt)	99 of 39,580	0.25%	[0.14%, 0.41%]	0.21%	[0.08%, 0.37%]
Q2	D2 class (stale receipt)	15 of 39,580	0.04%	[0.01%, 0.07%]	0.12%	[0.04%, 0.23%]
Q2	D3	45 of 39,580	0.11%	[0.03%, 0.25%]	0.02%	[0.01%, 0.03%]
Q2	D4	5,287 of 39,580	13.36%	[12.46%, 14.29%]	21.45%	[19.39%, 23.62%]
Q2	D7	5 of 39,580	0.01%	[0.00%, 0.03%]	0.03%	[0.00%, 0.10%]
Q2	at least one defect	5,391 of 39,580	13.62%	[12.71%, 14.56%]	21.74%	[19.69%, 23.90%]
Q3	D1 class (absent receipt)	317 of 42,763	0.74%	[0.45%, 1.09%]	0.24%	[0.07%, 0.56%]
Q3	D2 class (stale receipt)	13 of 42,763	0.03%	[0.00%, 0.08%]	0.01%	[0.00%, 0.03%]
Q3	D3	269 of 42,763	0.63%	[0.35%, 0.97%]	0.08%	[0.03%, 0.15%]
Q3	D4	3,571 of 42,763	8.35%	[7.10%, 9.78%]	28.36%	[23.47%, 33.40%]

stratum	defect class (failed predicate)	clearings	share	clustered 95%	share of value	clustered 95%
Q3	D7	2 of 42,763	0.00%	[0.00%, 0.01%]	0.00%	[0.00%, 0.00%]
Q3	at least one defect	3,829 of 42,763	8.95%	[7.69%, 10.39%]	28.48%	[23.60%, 33.49%]
Q4	D1 class (absent receipt)	80 of 34,371	0.23%	[0.17%, 0.30%]	0.11%	[0.06%, 0.18%]
Q4	D2 class (stale receipt)	28 of 34,371	0.08%	[0.00%, 0.25%]	0.05%	[0.00%, 0.14%]
Q4	D3	3 of 34,371	0.01%	[0.00%, 0.02%]	0.02%	[0.00%, 0.06%]
Q4	D4	2,243 of 34,371	6.53%	[5.97%, 7.15%]	7.50%	[6.37%, 8.70%]
Q4	D7	2 of 34,371	0.01%	[0.00%, 0.01%]	0.01%	[0.00%, 0.02%]
Q4	at least one defect	2,334 of 34,371	6.79%	[6.21%, 7.43%]	7.65%	[6.51%, 8.86%]

Appendix E: Consequence and gate effect by stratum

The H1 and H2 comparisons (Section 5.2) and the gate verdict against the corrective (Section 5.3), at W = 60 days, per stratum. Each cell gives the O rate with the corrective events and the clearings of its group; "none" marks an empty group. These are descriptive: the strata are reported beside the estimates and never modelled in. Company and vendor strata are in the slot files only, as hashed labels.

Item category: H1 and H2 (O rate, corrective events of clearings, at W = 60 days)

stratum	H1: D1/D2 failed	H1: three-way, all pass	H1 difference	H2: D4 failed	H2: D4 pass	H2 difference
2-way	none	none	n/a	0.00% (0 of 4)	2.72% (5 of 184)	-0.0272
3-way, invoice after receipt	none	4.68% (377 of 8,048)	n/a	6.81% (108 of 1,585)	4.69% (378 of 8,055)	0.0212
3-way, invoice before receipt	5.64% (39 of 692)	0.70% (909 of 129,554)	0.0493	1.41% (219 of 15,568)	0.73% (949 of 130,138)	0.0068

Item category: gate by outcome (clearings, W = 60 days)

stratum	analysed clearings	gate would have held	held, corrective within W	held, no corrective	allowed, corrective within W	allowed, no corrective
2-way	188	4 (2.13%)	0	4	5	179
3-way, invoice after receipt	9,640	1,592 (16.51%)	109	1,483	377	7,671
3-way, invoice before receipt	145,706	16,152 (11.09%)	259	15,893	909	128,645

Spend area: H1 and H2 (O rate, corrective events of clearings, at W = 60 days)

stratum	H1: D1/D2 failed	H1: three-way, all pass	H1 difference	H2: D4 failed	H2: D4 pass	H2 difference
Additives	7.69% (1 of 13)	1.02% (92 of 8,987)	0.0667	1.82% (22 of 1,207)	1.03% (93 of 8,999)	0.0079
CAPEX & SOCS	1.82% (1 of 55)	1.45% (63 of 4,331)	0.0036	2.16% (6 of 278)	1.46% (64 of 4,394)	0.0070

stratum	H1: D1/D2 failed	H1: three-way, all pass	H1 difference	H2: D4 failed	H2: D4 pass	H2 difference
Chemicals & Intermediates	none	0.00% (0 of 1)	n/a	none	0.00% (0 of 1)	n/a
Commodity Resins	0.00% (0 of 1)	2.22% (5 of 225)	-0.0222	1.25% (1 of 80)	2.21% (5 of 226)	-0.0096
Energy	none	none	n/a	none	0.00% (0 of 3)	n/a
Enterprise Services	0.00% (0 of 2)	2.05% (7 of 342)	-0.0205	0.00% (0 of 20)	2.00% (7 of 350)	-0.0200
Latex & Monomers	30.00% (3 of 10)	1.96% (41 of 2,096)	0.2804	1.88% (9 of 479)	2.09% (44 of 2,105)	-0.0021
Logistics	none	10.49% (205 of 1,955)	n/a	35.96% (41 of 114)	10.49% (205 of 1,955)	0.2548
Marketing	16.67% (1 of 6)	2.98% (33 of 1,106)	0.1368	12.04% (13 of 108)	3.06% (34 of 1,112)	0.0898
Others	0.00% (0 of 8)	1.44% (24 of 1,663)	-0.0144	5.81% (5 of 86)	1.54% (26 of 1,690)	0.0428
Packaging	5.29% (9 of 170)	0.45% (254 of 56,526)	0.0484	1.10% (76 of 6,901)	0.46% (263 of 56,660)	0.0064
Pigments & Colorants	40.00% (2 of 5)	1.22% (17 of 1,397)	0.3878	2.71% (8 of 295)	1.29% (18 of 1,399)	0.0143
Real Estate	none	0.00% (0 of 1)	n/a	0.00% (0 of 5)	2.05% (3 of 146)	-0.0205
Sales	13.57% (19 of 140)	0.73% (318 of 43,511)	0.1284	1.02% (33 of 3,244)	0.78% (340 of 43,647)	0.0024
Solvents	none	1.55% (21 of 1,354)	n/a	3.24% (7 of 216)	1.55% (21 of 1,357)	0.0169
Specialty Resins	none	1.03% (11 of 1,067)	n/a	3.51% (8 of 228)	1.03% (11 of 1,067)	0.0248
Spend Area Unidentified	none	0.00% (0 of 15)	n/a	0.00% (0 of 3)	0.00% (0 of 15)	0.0000
Titanium Dioxides	0.00% (0 of 5)	2.62% (8 of 305)	-0.0262	7.56% (9 of 119)	2.58% (8 of 310)	0.0498
Trading & End Products	0.73% (2 of 273)	1.22% (141 of 11,548)	-0.0049	2.13% (71 of 3,339)	1.22% (143 of 11,765)	0.0091
Workforce Services	none	16.89% (37 of 219)	n/a	17.74% (11 of 62)	16.89% (37 of 219)	0.0085
missing	25.00% (1 of 4)	0.94% (9 of 953)	0.2406	1.88% (7 of 373)	1.04% (10 of 957)	0.0083

Spend area: gate by outcome (clearings, W = 60 days)

stratum	analysed clearings	gate would have held	held, corrective within W	held, no corrective	allowed, corrective within W	allowed, no corrective
Additives	10,206	1,219 (11.94%)	23	1,196	92	8,895
CAPEX & SOCS	4,672	332 (7.11%)	7	325	63	4,277
Chemicals & Intermediates	1	0 (0.00%)	0	0	0	1
Commodity Resins	306	81 (26.47%)	1	80	5	220
Energy	3	0 (0.00%)	0	0	0	3
Enterprise Services	370	22 (5.95%)	0	22	7	341
Latex & Monomers	2,584	488 (18.89%)	12	476	41	2,055
Logistics	2,069	114 (5.51%)	41	73	205	1,750
Marketing	1,220	114 (9.34%)	14	100	33	1,073
Others	1,776	94 (5.29%)	5	89	26	1,656
Packaging	63,561	7,035 (11.07%)	85	6,950	254	56,272
Pigments & Colorants	1,694	297 (17.53%)	9	288	17	1,380
Real Estate	151	5 (3.31%)	0	5	3	143

stratum	analysed clearings	gate would have held	held, corrective within W	held, no corrective	allowed, corrective within W	allowed, no corrective
Sales	46,891	3,378 (7.20%)	55	3,323	318	43,195
Solvents	1,573	219 (13.92%)	7	212	21	1,333
Specialty Resins	1,295	228 (17.61%)	8	220	11	1,056
Spend Area Unidentified	18	3 (16.67%)	0	3	0	15
Titanium Dioxides	429	124 (28.90%)	9	115	8	297
Trading & End Products	15,104	3,556 (23.54%)	73	3,483	141	11,407
Workforce Services	281	62 (22.06%)	11	51	37	182
missing	1,330	377 (28.35%)	8	369	9	944

Month of the clearing: H1 and H2 (O rate, corrective events of clearings, at W = 60 days)

stratum	H1: D1/D2 failed	H1: three-way, all pass	H1 difference	H2: D4 failed	H2: D4 pass	H2 difference
2018-01	0.00% (0 of 1)	0.00% (0 of 238)	0.0000	0.00% (0 of 8)	0.00% (0 of 247)	0.0000
2018-02	0.00% (0 of 6)	0.30% (8 of 2,703)	-0.0030	2.08% (2 of 96)	0.29% (8 of 2,716)	0.0179
2018-03	13.04% (3 of 23)	0.88% (78 of 8,902)	0.1217	2.92% (25 of 855)	0.91% (81 of 8,935)	0.0202
2018-04	3.23% (3 of 93)	0.57% (76 of 13,223)	0.0265	0.92% (16 of 1,737)	0.59% (78 of 13,326)	0.0034
2018-05	4.41% (3 of 68)	0.82% (139 of 16,953)	0.0359	1.36% (26 of 1,909)	0.85% (144 of 17,030)	0.0052
2018-06	2.53% (2 of 79)	1.13% (197 of 17,400)	0.0140	2.12% (48 of 2,261)	1.14% (199 of 17,487)	0.0098
2018-07	12.00% (9 of 75)	1.20% (216 of 17,936)	0.1080	1.81% (36 of 1,991)	1.25% (225 of 18,028)	0.0056
2018-08	6.25% (6 of 96)	0.98% (186 of 19,054)	0.0527	1.80% (44 of 2,442)	1.00% (191 of 19,168)	0.0081
2018-09	2.56% (2 of 78)	0.87% (145 of 16,661)	0.0169	2.44% (55 of 2,253)	0.90% (151 of 16,749)	0.0154
2018-10	9.00% (9 of 100)	0.92% (153 of 16,683)	0.0808	2.01% (48 of 2,387)	0.98% (164 of 16,806)	0.0104
2018-11	2.74% (2 of 73)	1.12% (88 of 7,849)	0.0162	2.22% (27 of 1,218)	1.15% (91 of 7,885)	0.0106

Month of the clearing: gate by outcome (clearings, W = 60 days)

stratum	analysed clearings	gate would have held	held, corrective within W	held, no corrective	allowed, corrective within W	allowed, no corrective
2018-01	255	9 (3.53%)	0	9	0	246
2018-02	2,812	103 (3.66%)	2	101	8	2,701
2018-03	9,790	877 (8.96%)	28	849	78	8,835
2018-04	15,063	1,827 (12.13%)	18	1,809	76	13,160
2018-05	18,939	1,972 (10.41%)	29	1,943	141	16,826
2018-06	19,748	2,331 (11.80%)	50	2,281	197	17,220
2018-07	20,019	2,060 (10.29%)	45	2,015	216	17,743
2018-08	21,610	2,531 (11.71%)	49	2,482	186	18,893
2018-09	19,002	2,314 (12.18%)	58	2,256	148	16,540
2018-10	19,193	2,483 (12.94%)	59	2,424	153	16,557
2018-11	9,103	1,241 (13.63%)	30	1,211	88	7,774

Line value quartile: H1 and H2 (O rate, corrective events of clearings, at W = 60 days)

stratum	H1: D1/D2 failed	H1: three-way, all pass	H1 difference	H2: D4 failed	H2: D4 pass	H2 difference
Q1	2.59% (10 of 386)	0.56% (203 of 36,210)	0.0203	0.55% (11 of 2,012)	0.58% (212 of 36,537)	-0.0003
Q2	8.26% (9 of 109)	0.59% (210 of 35,733)	0.0767	0.61% (16 of 2,629)	0.61% (219 of 35,852)	-0.0000
Q3	5.75% (5 of 87)	0.86% (296 of 34,382)	0.0489	1.10% (49 of 4,447)	0.87% (301 of 34,516)	0.0023
Q4	13.64% (15 of 110)	1.84% (577 of 31,277)	0.1179	3.11% (251 of 8,069)	1.91% (600 of 31,472)	0.0120

Line value quartile: gate by outcome (clearings, W = 60 days)

stratum	analysed clearings	gate would have held	held, corrective within W	held, no corrective	allowed, corrective within W	allowed, no corrective
Q1	38,549	2,334 (6.05%)	20	2,314	203	36,012
Q2	38,481	2,724 (7.08%)	25	2,699	210	35,547
Q3	38,963	4,522 (11.61%)	54	4,468	296	34,145
Q4	39,541	8,168 (20.66%)	269	7,899	582	30,791

Vendor activity quartile: H1 and H2 (O rate, corrective events of clearings, at W = 60 days)

stratum	H1: D1/D2 failed	H1: three-way, all pass	H1 difference	H2: D4 failed	H2: D4 pass	H2 difference
Q1	11.43% (16 of 140)	2.08% (676 of 32,442)	0.0934	2.91% (176 of 6,056)	2.13% (699 of 32,764)	0.0077
Q2	14.91% (17 of 114)	0.76% (259 of 34,189)	0.1415	1.91% (101 of 5,287)	0.80% (276 of 34,293)	0.0111
Q3	1.21% (4 of 330)	0.72% (280 of 38,934)	0.0049	0.98% (35 of 3,571)	0.72% (284 of 39,192)	0.0026
Q4	1.85% (2 of 108)	0.22% (71 of 32,037)	0.0163	0.67% (15 of 2,243)	0.23% (73 of 32,128)	0.0044

Vendor activity quartile: gate by outcome (clearings, W = 60 days)

stratum	analysed clearings	gate would have held	held, corrective within W	held, no corrective	allowed, corrective within W	allowed, no corrective
Q1	38,820	6,194 (15.96%)	194	6,000	681	31,945
Q2	39,580	5,391 (13.62%)	118	5,273	259	33,930
Q3	42,763	3,829 (8.95%)	39	3,790	280	38,654
Q4	34,371	2,334 (6.79%)	17	2,317	71	31,966

Sources: `results/e2/confounders.json`, `results/e3/confounders.json`.